MASTERING UX/UI

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RESEARCH PLAN • UVA

Version of the Research Plan presented for the Mastering UX/UI Design course at EDIT, as a mandatory requirement.

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PRESENTATION

This material, prepared for evaluation, presents the research planning developed for the UVA software as part of the Mastering UX/UI course at EDIT. This project aimed to address the identified needs in restaurant management, with an emphasis on the integration and optimization of processes in the financial sector. Through an in-depth and structured study, the plan proposed innovative and practical solutions. Additionally, it details the objectives, methodologies, and stages that guided the development and validation of the system, ensuring its relevance and effectiveness.

SUMÁRIO

PRESENTATION 2

1 INTRODUCTION.....3

2 MIND MAP ……........3

3 STAKEHOLDER MAP 6

4. AREAS OF FOCUS 7

5. RESEARCH CANVAS.....8

6. PROTO-PERSONA 10

7. DESK RESEARCH …….......12

8. INTERVIEWS AND INSIGHTS.....16

9. PERSONAS 18

10. FROM USER JOURNEY TO FEATURES 20

11. CONCLUSION 23

INTRODUCTION

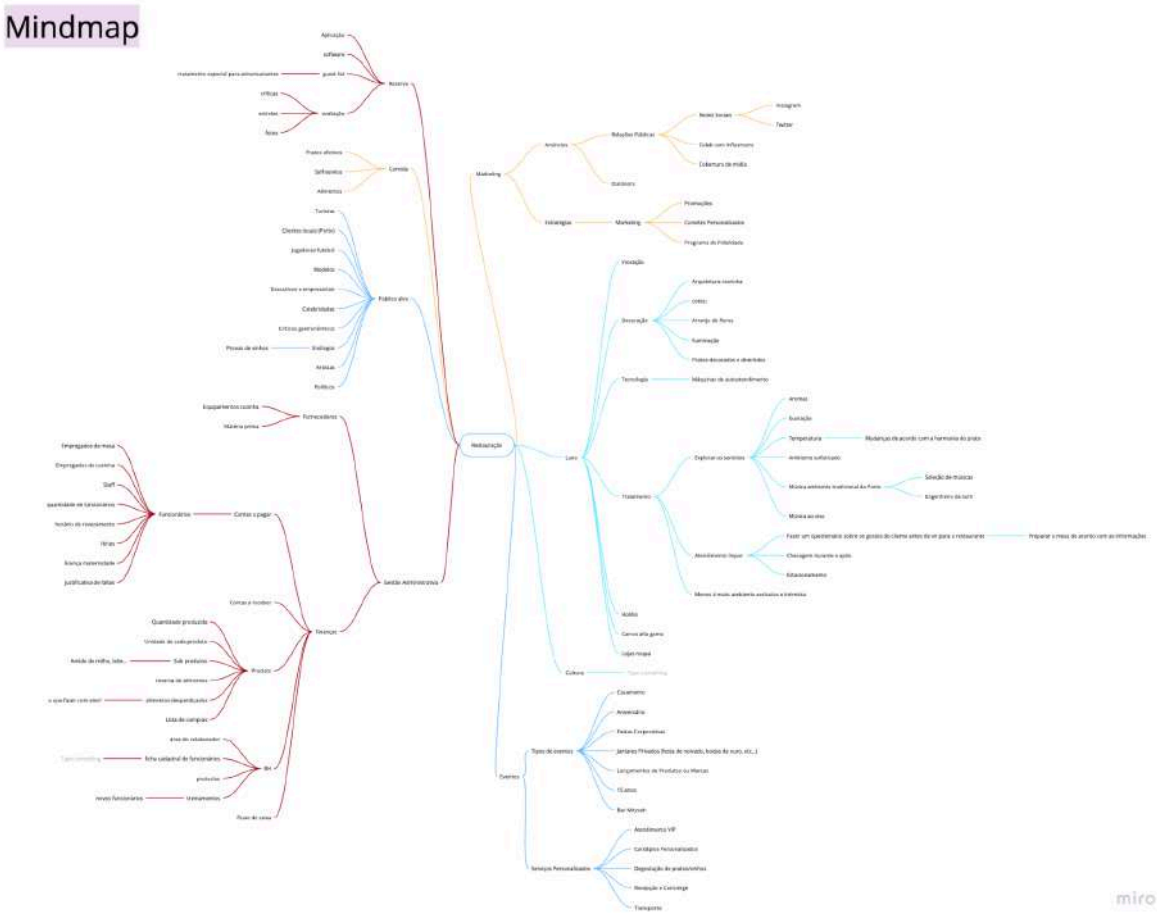
This Research Plan was developed with the objective of creating a centralized and integrated restaurant management software, named UVA, to address the specific

needs of the Alma Douro restaurant. The focus was on optimizing financial management, increasing operational efficiency, and offering personalized customer experiences, promoting the restaurant's growth and sustainability.

To structure the research, we followed four main stages. In the Understanding stage, we began by comprehending the restaurant's context and daily operations through interviews with managers and staff, capturing different perspectives. In the Planning stage, we planned how to address the identified issues, defining key questions and specific methodologies to collect valuable data. Next, in the Data Collection stage, we conducted interviews, on-site observations in the restaurant environment, and additional research to gather detailed information about the challenges faced. Finally, in the Validation stage, we validated the collected information, ensuring it reflected real needs and that the proposed solutions were effective. Each stage was carefully planned to ensure the approach remained consistent and oriented toward concrete results.

MIND MAP

Luxury restaurants, such as Alma Douro, aim to offer impeccable experiences to their customers. However, achieving this level of excellence involves facing significant operational challenges, which we addressed through the creation of a detailed mind map. This mind map was our initial tool to explore and organize the key areas related to the restaurant's operations, covering everything from administrative management to the customer experience.



Legend: Mind Map

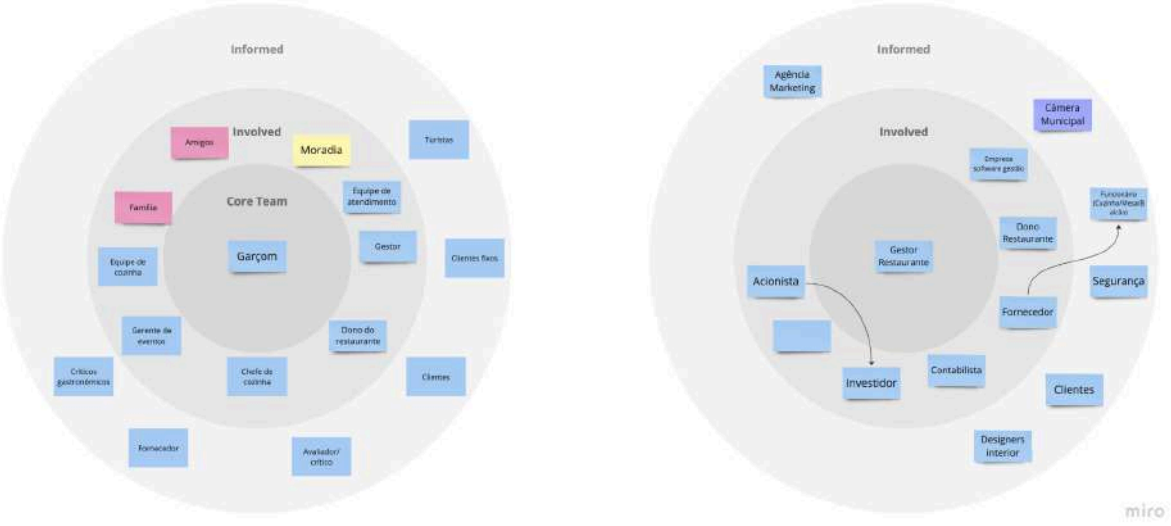
We identified essential categories such as Administrative Management, which included financial management, staff, and suppliers, fundamental for ensuring operational efficiency. In the Target Audience axis, we analyzed customer preferences, behaviors, and dietary needs, highlighting the importance of a personalized experience. In Food, we focused on the quality and consistency of dishes, while the Reservations theme brought to light the need for optimized and convenient processes for customers.

Additionally, in the Luxury category, we explored how to create a sophisticated environment that reflected the restaurant's values. In Marketing, we developed modern strategies, including social media and partnerships, to strengthen the brand. Finally, Culture and Events were highlighted as pillars to align the team's values and attract the target audience with memorable experiences.

The mind map provided a clear overview of the challenges and areas that required greater attention. From it, we connected operational issues to the processes involved and prioritized efforts on the most critical solutions for the restaurant's success.

STAKEHOLDER MAP

After organizing all the areas of impact in the restaurant through the mind map, the next step was to understand who the people and groups involved in these operations were. For this, we created a stakeholder map, which helped us identify the main actors and their relationships with the restaurant. This step was essential to ensure that the developed solutions considered all those involved in the process.



Legend: Stakeholder Map

In the Core Team, we identified the restaurant manager as the main nucleus, being primarily responsible for coordinating operations and making strategic decisions. In the group of Involved Stakeholders, we recognized those who frequently influenced or interacted with operations, such as suppliers, regular customers, and the restaurant owner. We considered their interactions to ensure the

system facilitated the necessary communication and support. Finally, in the group of Informed Stakeholders, we highlighted those who needed visibility or occasional reports, such as shareholders, marketing agencies, and even the City Council. These stakeholders did not participate directly in daily operations but were impacted by their results.

Mapping the stakeholders allowed us to identify all the people and groups with direct or indirect influence over the restaurant's operations. This vision was crucial for creating solutions that truly worked and met the needs of all.

This mapping provided clarity on who the system's users were, how they connected to the restaurant, and what their main expectations and needs were. With this foundation, we prioritized the product's functionalities strategically, ensuring they were directed toward the real demands of the different stakeholder groups.

AREAS OF FOCUS

With the construction of the mind map and stakeholder map, we discussed some initial ideas to guide possible solutions, recognizing that these were not the final point of the work but a starting point for development. Among the ideas, we highlighted the need for integrated financial management, optimization of restocking and inventory control, and specific functionalities for staff management, such as vacation and leave planning.

We also considered integrating software for order management and personalized reservations, although these were already partially covered by existing systems. We discussed the possibility of including strategic consulting tools and process automation through artificial intelligence, such as predictive notifications. Additionally, we analyzed ways to personalize task management with methodologies like Kanban and conducting continuous stakeholder interviews to refine solutions.

These ideas served as a starting point, providing valuable insights for the subsequent stages, but were understood as preliminary, subject to validation and refinement throughout the project. We knew the main focus lay in the financial sector,

given its strategic importance to the restaurant's operations, but we recognized the need to explore complementary areas such as inventory management, team management, and customer experience. This holistic approach ensured that the developed solutions considered integration between these pillars, promoting a more comprehensive and efficient vision for the software's success.

RESEARCH CANVAS

After defining the financial sector as the central axis for the project's development, we understood the need for a solid foundation to structure the research. For this, we used the Research Canvas as an essential tool in organizing and structuring the work. The canvas played a strategic role in guiding the process of clarifying objectives, formulating key questions, identifying stakeholders, and defining the most appropriate research methods.

The purpose of the research was to understand the specific needs and challenges faced by the restaurant manager in financial management. Our main objective was to explore how financial operations connected with other areas of the restaurant, creating a personalized solution to meet these managers' needs. To achieve this understanding, we formulated strategic questions addressing the main operational challenges. For example, we sought to understand which systems were currently used in financial management, what patterns and behaviors were repeated in the restaurant's daily routine, and how teams were organized. We also investigated the greatest difficulties in cost control and system integration, as well as the most relevant statistical data to support management decisions.

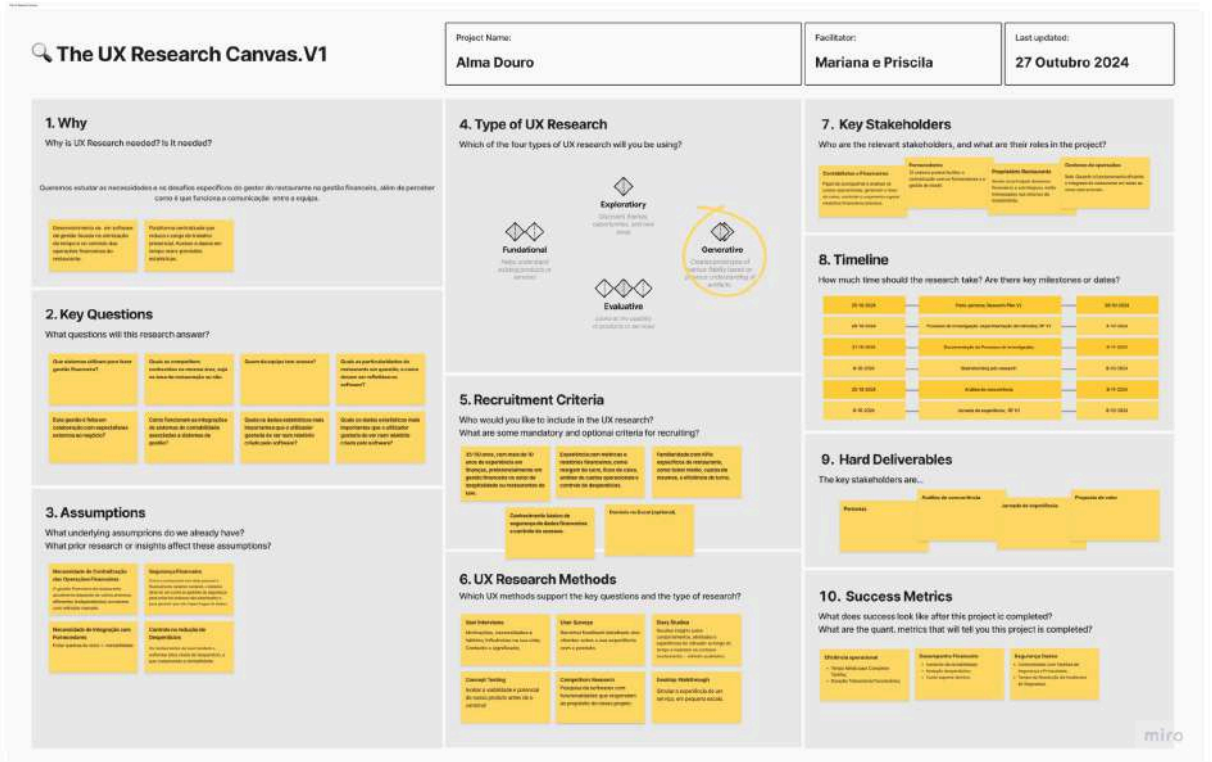
Based on some initial assumptions, we believed that managers faced difficulties in hiring and negotiating with reliable suppliers, in addition to relying on tools that offered financial security and detailed reports. We also identified the lack of integration between financial and operational systems as a factor generating significant inefficiencies. Finally, we highlighted the need to consolidate supplier and inventory data to optimize internal processes.

The methodological approach combined two types of UX research. We used generative research to explore new opportunities and identify unmet needs, while evaluative research played a dual role: testing and validating concepts directly with stakeholders and assessing the feasibility of the proposed ideas in the restaurant's real context.

Stakeholder mapping was fundamental to identifying the three main groups involved in the project. The core stakeholders included the restaurant manager, responsible for strategic decisions, while involved stakeholders encompassed suppliers, regular customers, and the operational team, who directly influenced daily operations. We also considered informed stakeholders, such as shareholders, marketing agencies, and regulatory authorities, who needed visibility into results, even if not actively participating in operations.

The research timeline was divided into distinct phases, allowing for organized and consistent progress. Between October 25 and 29, we created the Research Canvas and conducted desk research. From October 30 to November 2, we carried out stakeholder interviews and field observations. Between November 3 and 5, we performed competitor analysis and synthesized findings. Finally, from November 6 to 9, we held workshops and validated the results.

The project deliverables included personas representing the system's users, a journey map to illustrate operational flows, validated prototypes, and value propositions aligned with the identified needs. With this detailed structure and an approach focused on the real needs of restaurant managers, we believed the project would have a significant impact on optimizing financial operations and creating a more efficient and satisfying experience for all involved.



Legend: Research Canvas

PROTO-PERSONA

After constructing the Research Canvas, we created a proto-persona to consolidate and organize the information gathered during the initial research stage. Developing this proto-persona was essential to synthesize the main characteristics, needs, and behaviors observed in luxury restaurant managers, providing an initial representation to guide the next project stages. It served as an initial hypothesis to understand the user profile and would later be validated or adjusted based on data collected from direct interviews.

The proto-persona, named Miranda Prada, detailed the demographic aspects, habits, needs, objectives, behaviors, and obstacles faced by a luxury restaurant manager. In terms of habits and demographics, her background in Economics with a specialization in luxury restaurant management, international experience in France, and a lifestyle marked by an intense routine and self-description as a “workaholic”

stood out. Additionally, it was noted that the restaurant is located in a tourist region, which directly influences operational demands.

In terms of needs and objectives, the proto-persona revealed a desire for a platform that centralized functions often performed in person, reducing paperwork and increasing efficiency. Her objectives included having a low-cognitive-load solution, real-time access to operational data, and consolidating various management systems into a single integrated tool.

Her actions and behaviors showed a challenging routine, where she needed to manage her time between professional and personal responsibilities, often feeling overwhelmed. The lack of time was reflected in disorganized habits, such as delays, irregular eating, and difficulties delegating tasks, while she showed a preference for in-person meetings, highlighting her direct connection with the team.

Finally, the obstacles emphasized issues related to the lack of integrated tools, difficulties in monitoring operations, and concerns about using technologies that might not meet expectations or reduce empathy in team interactions. Challenges such as the distance between home and work and operational overload were also noted as factors impacting her routine and performance.

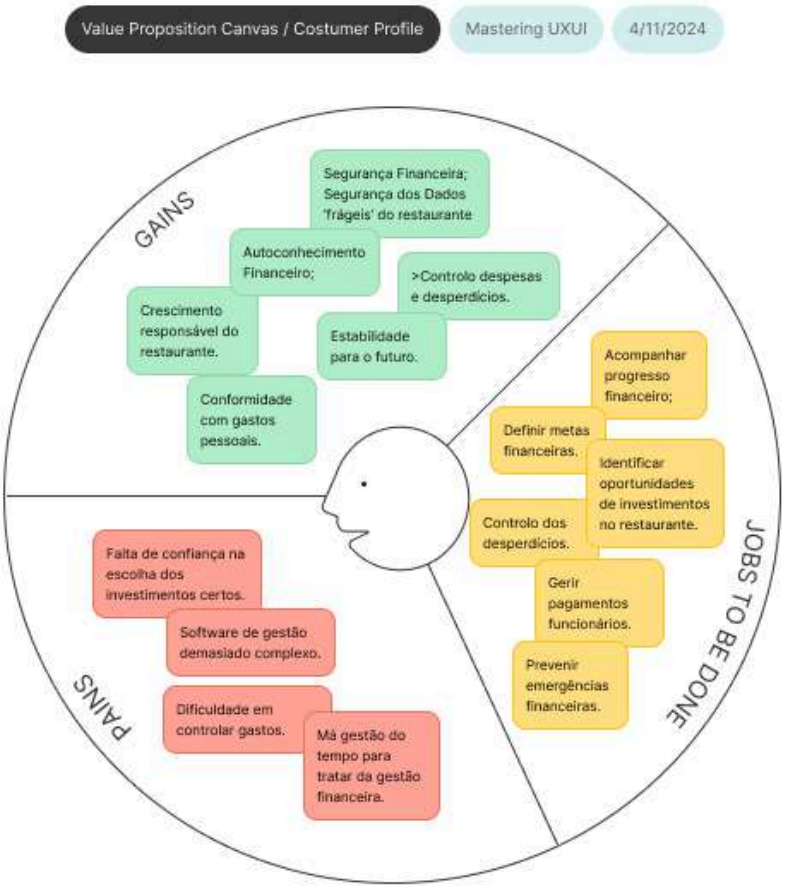
From the construction of this proto-persona, we moved to the next stages. The goal was to validate the raised hypotheses, allowing us to refine the observed points and gain a more detailed and realistic view of the needs and challenges faced in luxury restaurant management, ensuring the project remained aligned with users' real demands and that the proposed solutions were grounded in concrete data.

DESK RESEARCH

As mentioned earlier, we opted for a methodological approach that combines various research methods. We began with desk research, analyzing competitors' systems and industry benchmarks, and reviewing market reports on trends in luxury restaurant management. During this phase, we found revealing data, such as the fact that 50% of chefs and managers spend more than five hours weekly resolving issues

caused by non-integrated tools, and that 60% of restaurants consider their financial systems outdated, hindering the tracking of profits or losses.

During the Desk Research, we conducted the Value Proposition, which highlighted the needs and aspirations of restaurant managers regarding the financial and operational organization of their businesses. One of the main points observed was the expected benefits of adopting a solution, such as ensuring financial security and protecting the restaurant's strategic data. Additionally, the goal was to provide the client with greater self-awareness of the business's financial situation, enabling more effective control of expenses and waste. These gains reflected the desire for responsible growth, ensuring the restaurant could expand in a planned and sustainable manner without compromising financial stability. There was also a strong need to achieve a state of financial compliance and organization, both professionally and personally, fostering greater confidence in the future.



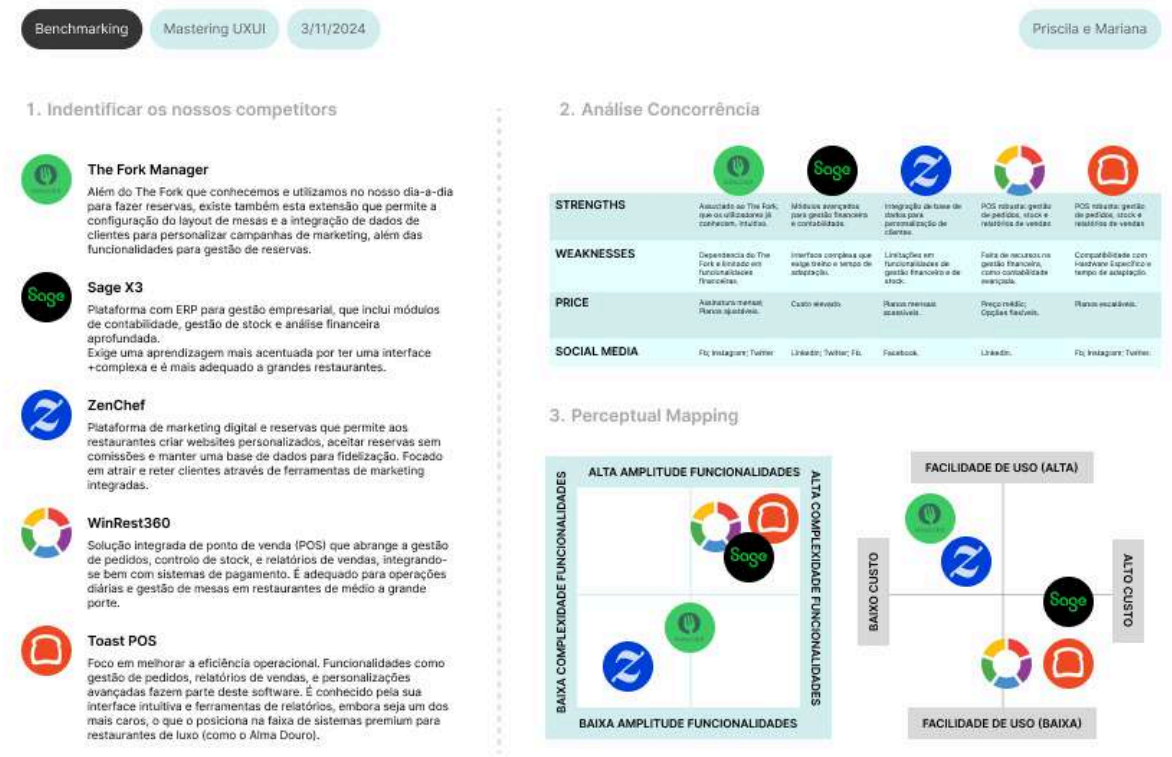
Legend: Value Proposition Canvas

On the other hand, the customer's pains were related to the difficulties faced in daily financial management. The lack of confidence in making investment decisions was one of the main points of frustration, particularly regarding resource allocation for improvements or restaurant expansion. Many management software solutions available on the market were complex and unintuitive, further complicating the manager's routine. Additionally, controlling expenses efficiently was a constant challenge, exacerbated by poor time management, which hindered a more strategic and organized approach to financial administration.

Based on this scenario, the customer's main tasks were identified as achieving control and clarity over the restaurant's financial situation. Continuous monitoring of financial progress, setting clear goals, and controlling waste were highlighted as priorities. Additionally, essential needs included managing staff payments, preventing financial emergencies, and identifying investment opportunities that supported sustainable business growth. These tasks expressed the manager's desire for a solution that simplified daily work and provided greater predictability for future decisions.

This context underpinned the research, which sought to deepen the understanding of managers' main difficulties and the solutions they expected to find. Through interviews and observation, it was possible to better understand customers' familiarity with financial tools and identify how they dealt, in practice, with cost control and planning challenges. The analysis also allowed us to identify gaps in available market solutions and direct the system's development toward these professionals' real needs, ensuring a proposal aligned with their expectations and goals for sustainable restaurant growth.

A Competitor Analysis was also conducted, aiming to identify the main market solutions addressing restaurant management demands, evaluating their strengths, weaknesses, prices, and social media presence. This allowed us to map the competitive landscape to identify opportunities for differentiation and improvements in the development of the system in question.



Legend: Competitor Analysis

Among the analyzed competitors, The Fork Manager stood out for its integration with widely used reservation tools, offering additional features such as table layout configuration and personalized marketing campaigns. Despite its strength in this segment, its dependence on The Fork was identified as a potential strategic limitation. On the other hand, Sage X3, a robust ERP platform, offered advanced accounting and financial management functionalities but presented a steeper learning curve, making it more suitable for large restaurants.

ZenChef focused on digital marketing and reservations, allowing restaurants to create personalized websites, accept commission-free reservations, and maintain a rich database for customer loyalty. However, its complexity in some aspects proved to be a barrier to adoption by smaller establishments. WinRest360 offered an integrated point-of-sale (POS) solution, covering everything from order control to

table management. Its accessibility made it a good option for mid-sized and large restaurants, although it could lack the personalization needed for high-end operations. Finally, Toast POS differentiated itself for its operational efficiency, with features tailored for premium restaurants, but at higher costs, positioning it as a niche solution for establishments like Alma D'Ouro.

In the perceptual map, it was possible to observe the competitors' positioning regarding the range of functionalities, complexity, and ease of use. While platforms like Sage X3 and Toast POS were situated in the high-complexity functional quadrant, solutions like The Fork Manager and ZenChef occupied more accessible positions. This contrast reinforced the need to find a balance between robust functionalities and ease of use, crucial for attracting both mid-sized restaurants and luxury operations. The analysis highlighted opportunities to develop a competitive solution with intuitive usability and functionalities adaptable to managers' specific needs.

INTEVIEWS AND INSIGHTS

Subsequently, we conducted qualitative interviews with 14 participants, including managers with over five years of experience in financial control and team coordination, chefs, and operational staff familiar with the dynamics of luxury restaurants, as well as regular customers who valued personalized experiences. We complemented the interviews with direct observations of employees' daily activities to identify inefficiencies.

After conducting the interviews with stakeholders and collecting qualitative data, we built a Research Wall to consolidate and synthesize the main research findings. This step was essential for identifying patterns, deepening the understanding of the needs and challenges faced, and structuring the information in a visual and collaborative way. During this analysis, we observed important characteristics that guided the next project stages, especially in the operational and managerial context of a small, intimate restaurant where quality and customer service were priorities.

The identified challenges centered on inventory management, financial control, and team administration. Despite the focus on service excellence and customer satisfaction, the restaurant faced difficulties in maintaining efficient processes and managing supplier relationships. One of the most relevant insights related to inventory management. The restaurant used the FIFO (First-In-First-Out) system to ensure older products were used first, and inventory levels were checked daily, with purchases planned according to need. However, we realized that manual control through spreadsheets limited efficiency and hindered waste and inventory tracking. Automating this process could significantly improve inventory control and reduce losses.

Another critical point identified was financial management, a vital part of the restaurant's operations. We observed constant monitoring of expenses and inventory, with monthly reports including costs, sales, and waste. Despite these efforts, the restaurant faced challenges with unexpected expenses and errors in inventory forecasting, negatively impacting financial stability. We also identified concerns with supplier contracts, which were regularly reviewed to address inflation impacts. These practices demonstrated the importance of a robust financial system to assist in planning and risk mitigation.

Team management and customer service also emerged as central elements. The restaurant stood out for creating a welcoming environment and dedicating attention to both customers and employees, promoting equitable treatment. However, staff management, especially in areas like cleaning, posed a significant challenge. Additionally, the owner's direct involvement, acting in both management and kitchen operations, reflected… *(text truncated in the original document)*.

Based on these insights, we made some recommendations to improve operational efficiency. Implementing an automated inventory management system was identified as a priority to reduce waste and optimize inventory tracking. Additionally, strengthening supplier relationships through more effective negotiation strategies could help control costs more efficiently. Finally, investing in team training and development was crucial to overcoming management challenges and enhancing service quality, reinforcing the restaurant's positioning as a space offering exceptional experiences.

This synthesis of the data obtained from interviews not only reinforced the initial learnings from the proto-persona but also provided additional nuances and depth to the understanding of the manager's context. It served as a solid foundation for defining the next steps, including solution validation and prototype development addressing the identified needs.

PERSONAS

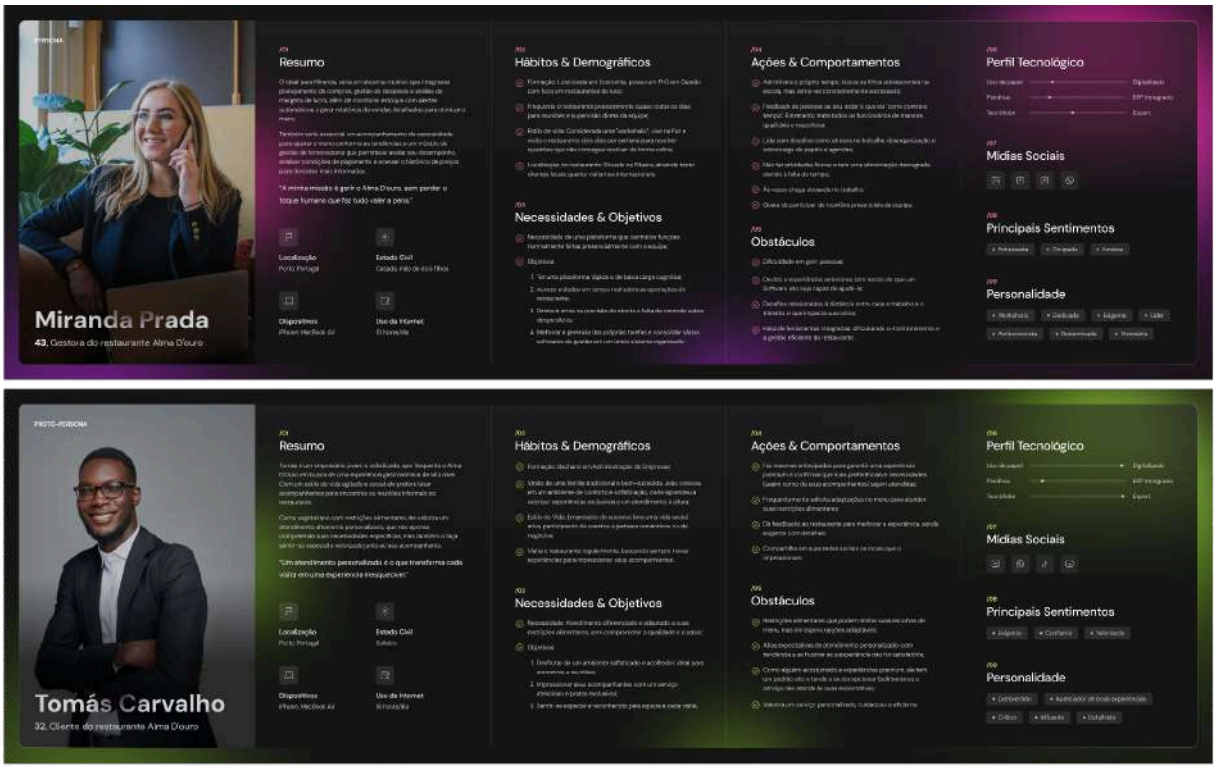
After conducting qualitative interviews and analyzing the insights obtained, we recognized the need to deepen the understanding of the two main profiles influencing the functioning and experience of the luxury restaurant Alma D'Ouro: the financial manager and the customer. From this information, we built two personas that synthesized the characteristics, needs, and challenges identified.

The first persona, Miranda Prada, represented the profile of the restaurant's financial manager. Based on data collected during interviews with experienced managers, her main challenges involved efficient cost management, waste control, and the need to consolidate administrative processes. Miranda was constructed as a highly skilled professional but overwhelmed by the restaurant's daily demands. Her needs included a desire for an integrated platform that simplified processes, provided real-time data, and optimized financial management. The insights also revealed difficulties with unintuitive tools, a lack of time for a strategic approach, and the importance of creating solutions promoting work-life balance.

The second persona, Tomás Carvalho, was developed to represent the typical Alma D'Ouro customer, inspired by interviews with regular customers, especially those valuing personalized and high-standard experiences. Tomás was characterized as a young, sophisticated, and demanding entrepreneur. The research revealed that customers of this profile sought exceptional service that met their personal preferences, such as dietary restrictions and vegetarian options, without compromising the quality or sophistication of the service. The interviews also highlighted the impact of an exclusive environment and the importance of tools facilitating reservations and personalized communication with the restaurant.

The construction of these two personas was fundamental for the project, as it provided a solid foundation for developing solutions directly addressing the needs of both profiles. For the financial manager, the proposed functionalities focused on system integration, task automation, and detailed reports to assist in strategic decision-making. For the customer, the ideas were oriented toward ensuring a personalized experience, from reservation to reflection, ensuring the service aligned with their expectations of exclusivity and attention to detail.

These personas not only synthesized the research results but also guided the subsequent project stages, ensuring the developed solutions were practical, applicable, and aligned with the real needs of the restaurant and its main stakeholders.



Legend: Personas

FROM USER JOURNEY TO FUNCTIONALITIES

After constructing the personas, we proceeded to create the User Journey, which allowed us to deepen our understanding of users' interactions with the Alma D'Ouro restaurant. Based on the characteristics, needs, and challenges identified in the personas, we mapped the key stages of the journey for both the customer and the manager, highlighting critical moments and opportunities for improvement. This process ensured that each proposed solution was directly aligned with the defined profiles, reinforcing the connection between the personas and the development of specific functionalities for the system.

Divided into phases such as discovery, reservation, restaurant interaction, and operational management, the User Journey mapping highlighted critical points and opportunities for improvement throughout the journey. Each stage was accompanied by an analysis of the user's emotions, actions, backstage operations, and insights for improvements.

For example, we identified that the reservation phase presented challenges in personalization and that the restaurant experience could be enriched with more integrated solutions. Additionally, the mapping included the managers' perspective, analyzing how processes like inventory control and financial analysis directly impact operations.

This stage was essential for mapping the main needs and challenges faced by users during their interaction with the Alma D'Ouro restaurant, covering both the managers' and customers' perspectives. From the stages mapped in the User Journey, we defined a List of Main Functionalities, aligned with the critical points and improvement opportunities identified throughout the process.

The backstage analysis in the User Journey revealed difficulties faced by managers in monitoring costs, waste, and profit margins. These findings led to the definition of crucial functionalities, such as cost and dish profitability management and detailed financial reports, providing real-time data and relevant insights to support more informed decision-making. In the reservation phase, we found the process to be complex and insufficiently personalized, which drove the creation of tools like the customer profile and history, integrating user preferences into

reservations, and the intelligent reservation system, automating table selection and significantly improving the customer experience.

During customer interaction and financial analysis, we identified the need to track dish performance. For this, we designed a functionality analyzing customer feedback and performance indicators, allowing managers to adjust the menu based on public preferences. Personalizing the customer experience, identified as a priority, led to tools like the personalized customer profile, preference-based notifications, and QR code check-in, providing a fluid and personalized experience from arrival to payment.

In the operational context, we observed the importance of reducing manual tasks and minimizing errors. To address this need, we designed functionalities like supplier management and automatic ingredient control, increasing operational efficiency and helping avoid waste.

Based on these data, we detailed functionalities that enrich and complement the system's main list, ensuring more efficient and tailored management for the restaurant's specific needs. Among the highlights is the dish analysis dashboard, which allows understanding the cost and value of each dish. This panel considers factors like ingredient weight and cost, as well as resource consumption such as electricity, water, gas, and equipment, offering managers a detailed view of each menu item's financial impact and supporting strategic decisions.

Another essential functionality is the automation of resource control, which automatically records and calculates water, electricity, and gas consumption per dish. This solution eliminates manual tasks, improves calculation accuracy, and helps identify inefficiencies, optimizing operational costs. Supplier integration was also highlighted as critical, allowing managers to easily access information like ingredient prices and availability. Additionally, any change in an ingredient's cost is automatically reflected in related dishes, ensuring consistency and efficiency.

The new dish value estimation functionality, supported by artificial intelligence, reduces the time needed to analyze recipes and facilitates the introduction of new items.

To complement, we developed an interactive tutorial functionality, guiding users in using the software's tools. This intuitive onboarding system simplifies the learning curve, especially for new employees or managers, ensuring an efficient and productive experience from the start.

These functionalities reflect the commitment to creating a robust system capable of meeting the restaurant's operational and financial needs while providing intuitive, practical, and tailored management for the business's reality. This Research Plan was crucial for grounding these solutions and guiding the creation of prototypes, ensuring each developed functionality accurately addressed the identified challenges and objectives.

CONCLUSION

This entire Research Plan was decisive in structuring the project and ensuring that the developed solutions were truly oriented toward the needs identified during the research process. The information collected and analyzed at each stage was crucial for creating the prototypes, enabling the development of a robust system aligned with the reality of the Alma D'Ouro restaurant. This detailed planning not only guided the design of the proposed solutions but also ensured they were practical, efficient, and effective in addressing the restaurant's challenges and objectives.